

Table 1: Supported datasets (pre-build, 2026-09-13) schematized + L0–L4 verified in the TorchCell database. *Instances* is the dataset length (total genotype×environment records); *Shape* the shape of one phenotype instance; *Graph role* where the label sits in the cell graph; *Signal (gzip, bits)* a Kolmogorov-complexity proxy – the gzip size in bits of the concatenated per-record stored instances (perturbation + environment + phenotype), so a natural isolate’s genome-defining perturbation set counts alongside its phenotype; the shared reference genome is never counted.

| Dataset                                    | Genotypes | Env   | Instances         | Phenotype                                             | Shape         | Graph role     | Signal (gzip, bits)  |
|--------------------------------------------|-----------|-------|-------------------|-------------------------------------------------------|---------------|----------------|----------------------|
| <b>Fitness + genetic interaction</b>       |           |       |                   |                                                       |               |                |                      |
| Costanzo 2016 smf                          | 20,484    | 2     | 20,484            | single-mutant fitness                                 | scalar        | global         | $5.4 \times 10^6$    |
| Costanzo 2016 dmf                          | 20.7M     | 2     | 20,705,612        | double-mutant fitness                                 | scalar        | global         | $7.1 \times 10^9$    |
| Costanzo 2016 dmi                          | 20.7M     | 2     | 20,705,612        | digenic interaction                                   | scalar        | edge           | $5.8 \times 10^9$    |
| Kuzmin 2018 smf                            | 1,539     | 1     | 1,539             | single-mutant fitness                                 | scalar        | global         | $3.5 \times 10^5$    |
| Kuzmin 2018 dmf                            | 410,399   | 1     | 410,399           | double-mutant fitness                                 | scalar        | global         | $1.5 \times 10^8$    |
| Kuzmin 2018 tmf                            | 91,111    | 1     | 91,111            | triple-mutant fitness                                 | scalar        | global         | $3.9 \times 10^7$    |
| Kuzmin 2018 dmi                            | 410,399   | 1     | 410,399           | digenic interaction                                   | scalar        | edge           | $1.3 \times 10^8$    |
| Kuzmin 2018 tmi                            | 91,111    | 1     | 91,111            | trigenic interaction                                  | scalar        | hyperedge      | $3.2 \times 10^7$    |
| Kuzmin 2020 smf                            | 472       | 1     | 472               | single-mutant fitness                                 | scalar        | global         | $9.1 \times 10^4$    |
| Kuzmin 2020 dmf                            | 632,797   | 1     | 632,797           | double-mutant fitness                                 | scalar        | global         | $2.3 \times 10^8$    |
| Kuzmin 2020 tmf                            | 301,798   | 1     | 301,798           | triple-mutant fitness                                 | scalar        | global         | $1.1 \times 10^8$    |
| Kuzmin 2020 dmi                            | 632,797   | 1     | 632,797           | digenic interaction                                   | scalar        | edge           | $2.0 \times 10^8$    |
| Kuzmin 2020 tmi                            | 301,798   | 1     | 301,798           | trigenic interaction                                  | scalar        | hyperedge      | $1.1 \times 10^8$    |
| Baryshnikova 2010 (smf)                    | 5,993     | 1     | 5,993             | single-mutant fitness                                 | scalar        | global         | $1.7 \times 10^6$    |
| O’Duibhir 2014 (smf)                       | 1,312     | 1     | 1,312             | single-mutant fitness                                 | scalar        | global         | $2.8 \times 10^5$    |
| <b>Environmental / chemogenomic</b>        |           |       |                   |                                                       |               |                |                      |
| Auesukaree 2009 (stress screen)            | 333       | 6     | 525               | stress sensitivity (categorical: sensitive/no_change) | scalar        | global         | $2.5 \times 10^5$    |
| Mota 2024 (weak-acid screen)               | 601       | 3     | 1,270             | weak-acid susceptibility (ordinal: 0/1/2 grades)      | scalar        | global         | $6.2 \times 10^5$    |
| Vanacloig-Pedros 2022                      | 3,647     | 41    | 143,218           | chemogenomic fitness (log2-ratio)                     | scalar        | global         | $8.7 \times 10^7$    |
| Costanzo 2021 (condition-SGA)              | 4,399     | 14    | 61,430            | differential mutant fitness                           | scalar        | global         | $1.0 \times 10^8$    |
| Hillenmeyer 2008 het (FitDb HIP)           | 5,814     | 514   | 2,698,797         | HIP fitness-defect log2-ratio                         | scalar        | global         | $8.1 \times 10^8$    |
| Hillenmeyer 2008 hom (FitDb HOP)           | 4,667     | 279   | 1,088,620         | HOP fitness-defect z-score                            | scalar        | global         | $3.4 \times 10^8$    |
| Wildenhain 2015 (drug tolerance)           | 256       | 5,168 | 428,206           | growth-inhibition z-score                             | scalar        | global         | $2.0 \times 10^8$    |
| Hoepfner 2014 (HIP/HOP atlas)              | 10,779    | 563   | 3,124,319         | HIP/HOP sensitivity score                             | scalar        | global         | $1.4 \times 10^9$    |
| Smith 2006 (chemogenomic)                  | 4,721     | 3     | 12,747            | chemogenomic sensitivity (clear-zone ordinal)         | scalar        | global         | $1.4 \times 10^6$    |
| Lian 2019 (MAGIC CRISPR-AID)               | 266,415   | 3     | 266,304           | furfural tolerance fitness (log2-ratio)               | scalar        | global         | $1.3 \times 10^8$    |
| Mormino 2022 (CRISPRi acetic-acid)         | 12        | 1     | 12                | acetic-acid sensitivity (categorical)                 | scalar        | global         | $1.2 \times 10^4$    |
| Smith 2016 (CRISPRi chem-genetic)          | 1,035     | 8     | 7,053             | chemogenomic fitness (log2-ratio)                     | scalar        | global         | $2.1 \times 10^6$    |
| Bloom 2019 (16-cross segregant panel)      | 13,950    | 38    | 530,100           | colony size residual / absolute                       | scalar        | global         | $2.5 \times 10^8$    |
| <b>Viability</b>                           |           |       |                   |                                                       |               |                |                      |
| SGD essentiality                           | 1,329     | 1     | 1,329             | gene essentiality                                     | scalar        | node           | $1.4 \times 10^5$    |
| SynLethDB (lethal)                         | 14,000    | 1     | 14,000            | synthetic lethality                                   | scalar        | edge           | $2.2 \times 10^6$    |
| SynLethDB (rescue)                         | 6,948     | 1     | 6,948             | synthetic rescue                                      | scalar        | edge           | $1.1 \times 10^6$    |
| <b>Morphology</b>                          |           |       |                   |                                                       |               |                |                      |
| Ohya 2005 (SCMD CalMorph)                  | 4,718     | 1     | 4,718             | cell morphology (CalMorph)                            | vector (281)  | global         | $1.5 \times 10^8$    |
| Ohnuki 2018 (SCMD CalMorph)                | 1,112     | 1     | 1,112             | cell morphology (CalMorph)                            | vector (281)  | global         | $4.8 \times 10^7$    |
| Ohnuki 2022 (SCMD CalMorph)                | 1,979     | 1     | 1,979             | cell morphology (CalMorph)                            | vector (281)  | global         | $8.8 \times 10^7$    |
| <b>Expression (microarray)</b>             |           |       |                   |                                                       |               |                |                      |
| Kemmeren 2014                              | 1,484     | 1     | 1,484             | mRNA log2(mut/wt)                                     | vector (6169) | node           | $3.2 \times 10^9$    |
| Sameith 2015 sm                            | 82        | 1     | 82                | mRNA log2(mut/ref)                                    | vector (6169) | node           | $1.8 \times 10^8$    |
| Sameith 2015 dm                            | 72        | 1     | 72                | mRNA log2(mut/ref)                                    | vector (6169) | node           | $1.7 \times 10^8$    |
| <b>Expression (RNA-seq)</b>                |           |       |                   |                                                       |               |                |                      |
| Caudal 2024 (pan-transcriptome)            | 943       | 1     | 943               | mRNA abundance (RNA-seq)                              | vector (6000) | node           | $1.5 \times 10^9$    |
| Nadal-Ribelles 2025 (Perturb-seq)          | 3,150     | 2     | 6,188             | mRNA logFC (Perturb-seq)                              | vector (5639) | node           | $2.2 \times 10^9$    |
| <b>Metabolite</b>                          |           |       |                   |                                                       |               |                |                      |
| Cachera 2023 (CRI-SPA betaxanthin)         | 4,735     | 1     | 4,735             | betaxanthin (product proxy)                           | scalar        | bipartite node | $2.2 \times 10^6$    |
| Müllerder 2016 (amino-acid metabolome)     | 4,678     | 1     | 4,678             | amino-acid concentrations                             | vector (19)   | bipartite node | $8.1 \times 10^6$    |
| Zelezniak 2018 (metabolome)                | 95        | 1     | 95                | metabolite levels                                     | vector (25)   | bipartite node | $3.1 \times 10^5$    |
| Ozaydin 2013 ( $\beta$ -carotene screen)   | 4,474     | 1     | 4,474             | $\beta$ -carotene (colony-color visual score)         | scalar        | global         | $1.0 \times 10^6$    |
| da Silveira 2014 (lipidomics)              | 127       | 1     | 127               | lipid-species relative abundance                      | vector (135)  | bipartite node | $1.0 \times 10^6$    |
| Yoshida 2012 (organic acids)               | 17        | 1     | 17                | organic-acid titer                                    | vector (6)    | bipartite node | $1.4 \times 10^4$    |
| Xue 2025 (free fatty acids, private)       | 176       | 1     | 176               | free-fatty-acid titer                                 | vector (5)    | bipartite node | $1.8 \times 10^5$    |
| Lopez 2024 (isobutanol screen, private)    | 4,554     | 1     | 4,554             | isobutanol biosensor fold-change                      | scalar        | bipartite node | $9.5 \times 10^5$    |
| Lopez 2024 (isobutanol validated, private) | 224       | 1     | 224               | isobutanol biosensor fold-change (validated)          | scalar        | bipartite node | $7.6 \times 10^4$    |
| <b>Protein abundance</b>                   |           |       |                   |                                                       |               |                |                      |
| Zelezniak 2018 (SWATH proteome)            | 97        | 1     | 97                | protein abundance                                     | vector (726)  | node           | $1.7 \times 10^7$    |
| Messner 2023 (proteome)                    | 4,699     | 1     | 4,699             | protein abundance                                     | vector (1830) | node           | $1.1 \times 10^9$    |
| <b>Total (50 datasets)</b>                 |           |       | <b>52,738,566</b> |                                                       |               |                | $2.6 \times 10^{10}$ |

In scope but not yet built/verified (no signal): Ho 2009 (bioethanol tolerance – genotype needs a WGS pipeline); Trika 2015 (sclareol titers – figure-only data).